

# 1.LangSmith的使用

LangChain生态系统中专门用于LLM应用**调试、监控、评估和管理**的平台

- **调试(tracing)**:记录每次LLM调用的详细信息
- **监控(monitors)**:实施查看应用性能
- **调试(debug)**:排查问题和优化性能
- **评估(evaluate)**:系统化测试LLM应用

langsmith和langchain因为是同一生态，只要配置好env文件：

LANGCHAIN\_TRACING\_V2=true

LANGCHAIN\_API\_KEY=ls\_XXXXXXXXXXXXXXXXXXXX

LANGCHAIN\_PROJECT=test\_project

LANGCHAIN\_ENDPOINT=<https://api.smith.langchain.com>

代码中无需任何操作

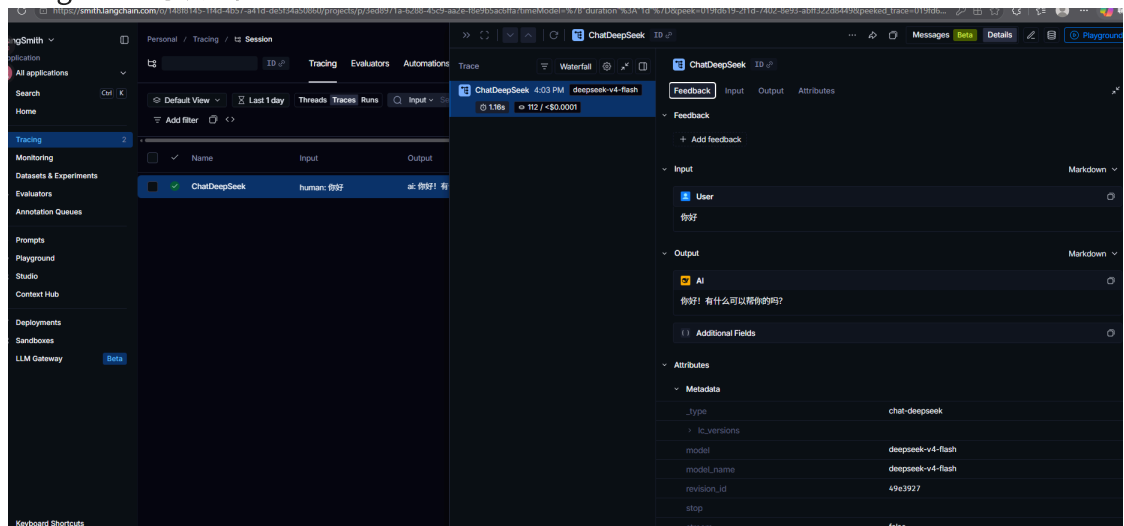
```
In [1]: from dotenv import load_dotenv
        from langchain_deepseek import ChatDeepSeek
        import os

        load_dotenv(override=True) # 表示.env文件优先
        # 强制开启追踪，兜底防止.env没加载到
        # os.environ["LANGCHAIN_TRACING_V2"] = "true"
        model = ChatDeepSeek(
            api_key=os.getenv("DEEPSEEK_API_KEY"),
            base_url=os.getenv("DEEPSEEK_BASE_URL"),
            model="deepseek-v4-flash"
        )

        print(model.invoke("你好"))
```

```
content='你好！有什么可以帮你的吗？' additional_kwargs={'refusal': None, 'reasoning_content':
'We need respond in Chinese. Need maybe greeting. User says 你好. Need answer friendly.'}
response_metadata={'token_usage': {'completion_tokens': 28, 'prompt_tokens': 84,
'total_tokens': 112, 'completion_tokens_details': {'accepted_prediction_tokens': None,
'audio_tokens': None, 'reasoning_tokens': 19, 'rejected_prediction_tokens': None},
'prompt_tokens_details': {'audio_tokens': None, 'cached_tokens': 0},
'prompt_cache_hit_tokens': 0, 'prompt_cache_miss_tokens': 84}, 'model_provider': 'deepseek',
'model_name': 'deepseek-v4-flash', 'system_fingerprint':
'fp_a18b46594c_prod0820_fp8_kvcache_20260402', 'id': '1a67ed8d-10b3-45ef-921d-
10a18696ae7f', 'finish_reason': 'stop', 'logprobs': None} id='lc_run--019fd619-2f1d-7402-8e93-
abff322d8449-0' tool_calls=[] invalid_tool_calls=[] usage_metadata={'input_tokens': 84,
'output_tokens': 28, 'total_tokens': 112, 'input_token_details': {'cache_read': 0},
'output_token_details': {'reasoning': 19}}
```

## LangSmith可以监控



In [2]: **from** langchain\_openai **import** ChatOpenAI  
**from** dotenv **import** load\_dotenv  
**import** os

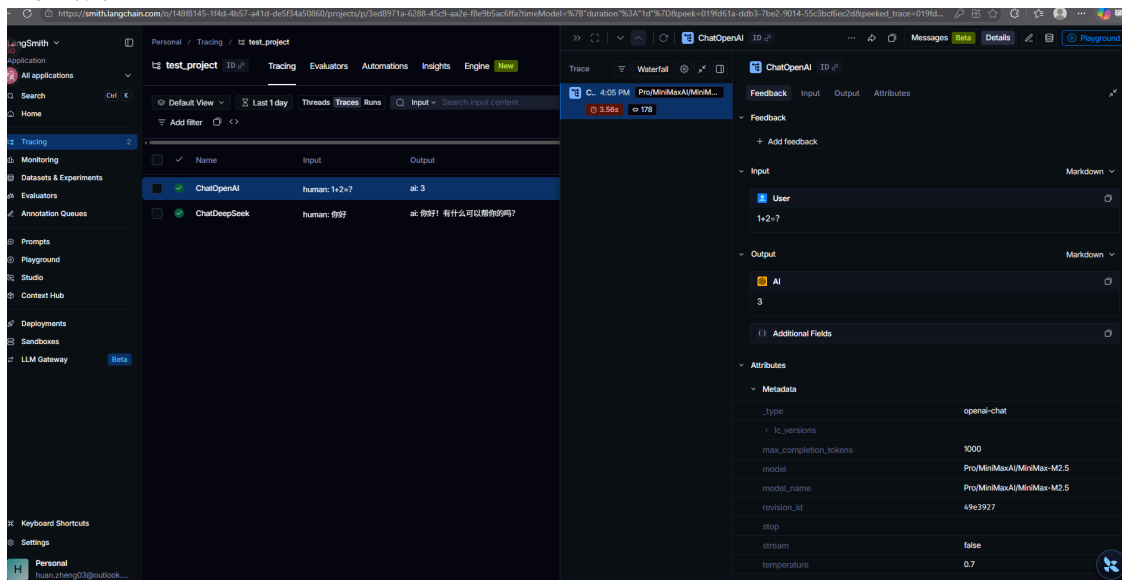
# 从 .env 文件中加载环境变量  
load\_dotenv(override=**True**)

```
model = ChatOpenAI(
    model="Pro/MiniMaxAI/MiniMax-M2.5",
    temperature=0.7,
    max_tokens=1000,
    max_retries=6,
    api_key=os.getenv("SILICON_API_KEY"),
    base_url=os.getenv("SILICON_BASE_URL")
)
```

```
print(model.invoke("1+2= ?"))
```

```
content='\n\n3' additional_kwargs={'refusal': None} response_metadata={'token_usage':
{'completion_tokens': 135, 'prompt_tokens': 43, 'total_tokens': 178, 'completion_tokens_details':
{'accepted_prediction_tokens': None, 'audio_tokens': None, 'reasoning_tokens': 133,
'rejected_prediction_tokens': None}, 'prompt_tokens_details': {'audio_tokens': None,
'cached_tokens': 0, 'prompt_cache_hit_tokens': 0, 'prompt_cache_miss_tokens': 43},
'model_provider': 'openai', 'model_name': 'Pro/MiniMaxAI/MiniMax-M2.5', 'system_fingerprint': '',
'id': '019fd61ada14ec2302206990e1ecd6e5', 'finish_reason': 'stop', 'logprobs': None} id='lc_run-
-019fd61a-ddb3-7be2-9014-55c3bcf6ec2d-0' tool_calls=[] invalid_tool_calls=[] usage_metadata=
{'input_tokens': 43, 'output_tokens': 135, 'total_tokens': 178, 'input_token_details':
{'cache_read': 0, 'output_token_details': {'reasoning': 133}}
```

监控结果：



```
In [3]: import os
from dotenv import load_dotenv
from langchain.chat_models import init_chat_model
from rich import print as rprint

load_dotenv()

llm = init_chat_model(
    model="zai-org/GLM-5.2",
    model_provider="openai",
    api_key=os.getenv("SILICON_API_KEY"),
    base_url=os.getenv("SILICON_BASE_URL"),
    temperature=0.5,
    max_tokens=200,
    # 指定可调整的参数
    configurable_fields=["model","model_provider","temperature","max_tokens"],
)

config = {
    "run_name": "joke_generation", # 在LangSmith中这次运行会显示为"joke_generation"
    "tags": ["my_tag1", "my_tag2"], # 打上标签便于分类查找
    "metadata": {
        "user_id": "victorzh", # 记录用户ID
        "session_id": "sess_123" # 记录会话ID
    },
    "configurable": {
        "model": "deepseek-ai/DeepSeek-V4-Flash", # 配置模型参数
        "model_provider": "deepseek", # 配置模型提供商
        "temperature": 0.7, # 配置温度参数
        "max_tokens": 1000 # 配置最大令牌数
    }
}

# 调用模型并传入config
res = llm.invoke("2*5=?", config=config)
rprint(res)
```

```

AIMessage(
  content='2*5=10',
  additional_kwargs={
    'refusal': None,
    'reasoning_content': 'We are asked: "2*5=?" This is a
simple multiplication problem. The answer is 10. But
the user wrote it in Chinese? Actually it\'s just "2*5=?". So
answer: 10.'
  },
  response_metadata={
    'token_usage': {
      'completion_tokens': 55,
      'prompt_tokens': 88,
      'total_tokens': 143,
      'completion_tokens_details': {
        'accepted_prediction_tokens': None,
        'audio_tokens': None,
        'reasoning_tokens': 50,
        'rejected_prediction_tokens': None
      },
      'prompt_tokens_details': {'audio_tokens': None,
'cached_tokens': 0},
      'prompt_cache_hit_tokens': 0,
      'prompt_cache_miss_tokens': 88
    },
    'model_provider': 'deepseek',
    'model_name': 'deepseek-ai/DeepSeek-V4-Flash',
    'system_fingerprint': '',
    'id': '019fd61bfc81215fd4fcd7dd4fe5927b',
    'finish_reason': 'stop',
    'logprobs': None
  },
  id='lc_run--019fd61c-0075-7330-b4cf-25d10e2f8a29-0',
  tool_calls=[],
  invalid_tool_calls=[],
  usage_metadata={
    'input_tokens': 88,
    'output_tokens': 55,
    'total_tokens': 143,
    'input_token_details': {'cache_read': 0},
    'output_token_details': {'reasoning': 50}
  }
)

```

监控结果：

The screenshot displays the LangSmith interface for monitoring LLM runs. The left sidebar contains navigation links for various features. The central panel shows a table of runs under the 'test\_project' workspace. The 'joke\_generation' run is highlighted, showing its input and output. The right panel provides a detailed view of the selected run, including its metadata and usage statistics.

| Run Name        | Input        | Output           |
|-----------------|--------------|------------------|
| joke_generation | human: 2*5=7 | ai: 2*5=10       |
| ChatOpenAI      | human: 1+2=7 | ai: 3            |
| ChatDeepSeek    | human: 你好    | ai: 你好！有什么可以帮你的吗 |

**Run Details for 'joke\_generation':**

- Model:** deepseek-ai/DeepSeek-V4-Flash
- Session ID:** 69c3927
- Usage:** 143 total tokens (88 input, 55 output)
- Metadata:** chat-deepseek, type: chat-deepseek, max\_tokens: 1000, temperature: 0.7

## 2.Langfuse本地化部署及使用

### 1) Langfuse本地部署

使用docekr部署

```
In [ ]: # git 拉取项目到本地
git clone https://github.com/langfuse/langfuse.git
# 切换到项目文件路径
cd langfuse
# docker compose启动
docker compose up
```

注意本地部署有可能端口冲突

到docker compose.yaml文件中修改端口保存

```
In [ ]: docker-compose down

docker compoes up -d
```

就可以通过

<http://localhost:3000>登录langfuse

langfuse的setting可以创建APIkey

配置到项目.env文件

env配置：

### LANGFUSE配置

LANGFUSE\_SECRET\_KEY="sk-lf-8021cd20-562f-40ac-a04f-89bdcc29e1cf"

LANGFUSE\_PUBLIC\_KEY="pk-lf-9e096ed5-9e84-4d3d-ab96-f6823c8366e4"

LANGFUSE\_BASE\_URL="<http://localhost:3000>"

代码中只要加入下面的设置即可：

```
In [ ]: # 1. 初始化Langfuse，自动读.env变量
langfuse = Langfuse()
# 生成LangChain回调处理器
lf_handler = CallbackHandler()

# 模型初始化需要在config参数加入参数
"callbacks":[lf_handler]
```

### 2) 代码tracing举例

```
In [1]: import os
from dotenv import load_dotenv
from langchain.chat_models import init_chat_model
```

```

from rich import print as rprint
from langfuse import Langfuse
from langfuse.langchain import CallbackHandler

load_dotenv(r"C:\Users\huan.zheng\myproject\LangChainLearning\.env")

# 1. 初始化Langfuse, 自动读.env变量
langfuse = Langfuse()
# 生成LangChain回调处理器
lf_handler = CallbackHandler()

llm = init_chat_model(
    model="zai-org/GLM-5.2",
    model_provider="openai",
    api_key=os.getenv("SILICON_API_KEY"),
    base_url=os.getenv("SILICON_BASE_URL"),
    temperature=0.5,
    max_tokens=200,
    # 指定可调整的参数
    configurable_fields=["model","model_provider","temperature","max_tokens"],
)

config = {
    "run_name": "calculate", # 在Langfuse中这次运行会显示为"calculate"
    "tags": ["my_tag1", "my_tag2"], # 打上标签便于分类查找
    "metadata": {
        "user_id": "victorzh-01", # 记录用户ID
        "session_id": "sess_124" # 记录会话ID
    },
    "configurable": {
        "model": "deepseek-ai/DeepSeek-V4-Flash", # 配置模型参数
        "model_provider": "deepseek", # 配置模型提供商
        "temperature": 0.7, # 配置温度参数
        "max_tokens": 1000 # 配置最大令牌数
    },
    "callbacks": [lf_handler] # 在配置中加入langfuse连接
}

# 调用模型并传入config
res = llm.invoke("5*800=?", config=config)
rprint(res)

```

c:\Users\huan.zheng\AppData\Local\miniconda3\envs\langchain1\Lib\site-packages\langgraph\checkpoint\base\\_\_init\_\_.py:17: LangChainPendingDeprecationWarning: The default value of `allowed\_objects` will change in a future version. Pass an explicit value (e.g., allowed\_objects='messages' or allowed\_objects='core') to suppress this warning.  
from langgraph.checkpoint.serde.jsonplus import JsonPlusSerializer

```

AIMessage(
  content='5乘以800等于4000。',
  additional_kwargs={
    'refusal': None,
    'reasoning_content': '我们被问到："5*800=？"
这是一个简单的乘法问题。5乘以800等于4000。所以答案应该是4000。但用户可能期望一个直接的答案。由于这是一个中文问题，
我们可以用中文回答。但作为AI，我们应该给出正确的数学答案。所以回答：4000。'
  },
  response_metadata={
    'token_usage': {
      'completion_tokens': 71,
      'prompt_tokens': 88,
      'total_tokens': 159,
      'completion_tokens_details': {
        'accepted_prediction_tokens': None,
        'audio_tokens': None,
        'reasoning_tokens': 64,
        'rejected_prediction_tokens': None
      },
      'prompt_tokens_details': {'audio_tokens': None},
    },
    'cached_tokens': 0,
    'prompt_cache_hit_tokens': 0,
    'prompt_cache_miss_tokens': 88
  },
  'model_provider': 'deepseek',
  'model_name': 'deepseek-ai/DeepSeek-V4-Flash',
  'system_fingerprint': '',
  'id': '019fd6273b41ca8e6de76c9dbeda3862',
  'finish_reason': 'stop',
  'logprobs': None
),
id='lc_run--019fd627-3f32-7aa3-87a8-7ac75a2e8be4-0',
tool_calls=[],
invalid_tool_calls=[],
usage_metadata={
  'input_tokens': 88,
  'output_tokens': 71,
  'total_tokens': 159,
  'input_token_details': {'cache_read': 0},
  'output_token_details': {'reasoning': 64}
}
)

```

结果：

